

Branch- Common to All Discipline

New Scheme Based On AICTE Flexible Curricula

BT301	Mathematics-III	3L-1T-0P	4 Credits
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OBJECTIVES: The objective of this course is to fulfill the needs of engineers to understand applications of Numerical Analysis, Transform Calculus and Statistical techniques in order to acquire mathematical knowledge and to solving wide range of practical problems appearing in different sections of science and engineering. More precisely, the objectives are:

- To introduce effective mathematical tools for the Numerical Solutions algebraic and transcendental equations.
- To enable young technocrats to acquire mathematical knowledge to understand Laplace transformation, Inverse Laplace transformation and Fourier Transform which are used in various branches of engineering.
- To acquaint the student with mathematical tools available in Statistics needed in various field of science and engineering.

Module 1: Numerical Methods – 1: (8 hours): Solution of polynomial and transcendental equations – Bisection method, Newton-Raphson method and Regula-Falsi method. Finite differences, Relation between operators, Interpolation using Newton's forward and backward difference formulae. Interpolation with unequal intervals: Newton's divided difference and Lagrange's formulae.

Module 2: Numerical Methods – 2: (6 hours): Numerical Differentiation, Numerical integration: Trapezoidal rule and Simpson's 1/3rd and 3/8 rules. Solution of Simultaneous Linear Algebraic Equations by Gauss's Elimination, Gauss's Jordan, Crout's methods, Jacobi's, Gauss-Seidal, and Relaxation method.,

Module 3: Numerical Methods – 3: (10 hours): Ordinary differential equations: Taylor's series, Euler and modified Euler's methods. RungeKutta method of fourth order for solving first and second order equations. Milne's and Adam's predictor-corrector methods. Partial differential equations: Finite difference solution two dimensional Laplace equation and Poission equation, Implicit and explicit methods for one dimensional heat equation (Bender-Schmidt and Crank-Nicholson methods), Finite difference explicit method for wave equation.

Module 4: Transform Calculus: (8 hours): Laplace Transform, Properties of Laplace Transform, Laplace transform of periodic functions. Finding inverse Laplace transform by different methods, convolution theorem. Evaluation of integrals by Laplace transform, solving ODEs by Laplace Transform method, Fourier transforms.

Module 5: Concept of Probability: (8 hours): Probability Mass function, Probability Density Function, Discrete Distribution: Binomial, Poisson's, Continuous Distribution: Normal Distribution, Exponential Distribution.

Textbooks/References:

1. P. Kandasamy, K. Thilagavathy, K. Gunavathi, Numerical Methods, S. Chand & Company, 2nd Edition, Reprint 2012.
2. S.S. Sastry, Introductory methods of numerical analysis, PHI, 4th Edition, 2005.
3. Erwin kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
4. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 35th Edition, 2010.
5. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2010.
6. Veerarajan T., Engineering Mathematics, Tata McGraw-Hill, New Delhi, 2008.
7. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal Book Stall, 2003 (Reprint).
8. S. Ross, A First Course in Probability, 6th Ed., Pearson Education India, 2002.
9. W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, 3rd Ed., Wiley, 1968. Statistics

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

New Scheme Based On AICTE Flexible Curricula

Aircraft Maintenance Engineering, III Semester

AF 302 - ELEMENTS OF AERONAUTICS

GOAL

To introduce the basic concepts of aerospace engineering and the current developments in the field.

OBJECTIVES

The course should enable the student to :

1. Understand the Historical evaluation of Airplanes
2. Study the different component systems and functions
3. Understand the basic properties and principles behind the flight
4. Study the different structures & construction
5. Study the various types of power plants used in aircrafts

OUTCOME

The student should be able to understand :

1. The history of aircraft & developments over the years
2. The types & classifications of components and control systems
3. The basic concepts of flight & Physical properties of Atmosphere
4. The types of fuselage and constructions and Landing gear systems
5. Different types of Engines and principles of Rocket

UNIT I HISTORICAL EVALUATION

Early airplanes, biplanes and monoplanes, Developments in aerodynamics, materials, Structures and propulsion over the years.

UNIT II AIRCRAFT CONFIGURATIONS

Components of an airplane and their functions. Different types of flight vehicles, classifications.

Conventional control, Powered control, Basic instruments for flying, typical systems for control Actuation.

UNIT III INTRODUCTION TO PRINCIPLES OF FLIGHT

Physical properties and structure of the atmosphere, Temperature, pressure and altitude Relationships, Evolution of lift, drag and moment. Aerofoils, Mach number, Manoeuvres.

UNIT IV INTRODUCTION TO AIRPLANE STRUCTURES

General types of construction, Monocoque, semi-monocoque and geodesic construction, Typical wing and fuselage structure. Landing Gear Structure

UNIT V POWER PLANTS USED IN AIRPLANES

Basic ideas about piston, turboprop and jet engines, Use of propeller and jets for thrust Production. Comparative merits, Principles of operation of rocket, types of rockets and typical Applications, Exploration into space.

TEXT BOOK

1. Anderson, J.D., "Introduction to Flight", McGraw-Hill, 1995.

REFERENCE

1. Kermode, A.C., "Flight without Formulae", McGraw-Hill, 1997.

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New Scheme Based On AICTE Flexible Curricula

Aircraft Maintenance Engineering, III Semester

AF 303 - AERO ENGINEERING THERMODYNAMICS

GOAL

To give a brief background of application of various laws of thermodynamics and its application in heat transfer, refrigeration and air-conditioning, jet propulsion system.

OBJECTIVES

The course should enable the students to :

1. Have a basic idea about Thermodynamic Systems, and processes.
2. Understand the air cycles like (Otto, Diesel, Dual combustion and Brayton combustion cycles) They should understand PV diagrams of four stroke and two stroke IC Engines.
3. Understand the thermodynamics of One Dimensional fluid flow and the application of Continuity and energy equations Properties of steam .To understand the Simple jet propulsion system and Thrust rocket motor
4. Understand about the refrigeration and Principles of Air conditioning and understand the Coefficient of performance and Properties of refrigerants.

OUTCOME

The students should be able to :

1. Understand the basic thermodynamic systems.
2. Understanding about the air cycles, and understanding about the plot of the PV diagrams of four stroke and two stroke IC Engines
3. Understand about the One Dimensional fluid flow and the applications of the Continuity equation and understand about the simple jet propulsion systems.
4. Understand about the Principles of refrigeration and Air conditioning and understand the Coefficient of performance and Properties of refrigerants.

UNIT I BASIC THERMODYNAMICS

Systems, Zeroth Law, First Law - Heat and work transfer in flow and non-flow processes, Second law, Kelvin- Planck statement - Clausius statement - concept of entropy - Clausius inequality - entropy change in non-flow processes.

UNIT II AIR CYCLES

Otto, Diesel, Dual combustion and Brayton combustion cycles - Air standard efficiency - Mean effective pressure - Actual and theoretical PV diagrams of four stroke and two stroke IC Engines.

UNIT III THERMODYNAMICS OF ONE DIMENSIONAL FLUID FLOW

Application of Continuity and energy equations- Properties of steam - Rankine cycle - Isentropic flow of ideal gases through nozzles - Simple jet propulsion system - Thrust rocket motor - Specific impulse.

UNIT IV REFRIGERATION AND AIR CONDITIONING

Principles of refrigeration, Air conditioning - Heat pumps - Vapour compression - Vapour absorption types - Coefficient of performance, Properties of refrigerants.

UNIT V AIR COMPRESSORS

Classification and working principle, work of compression with and without clearance volume, Isothermal and Isentropic efficiency of reciprocating air compressors, multistage compression and intercooling. Various types of compressors (Descriptive treatment only).

TEXT BOOKS

1. Rathakrishnan, E, "Fundamentals of Engineering Thermodynamics", Prentice - Hall, India, 2000
2. Nag, P.K., "Engineering Thermodynamics", Tata McGraw-Hills Co., Ltd., Seventh Edn., 1993
3. Yunus A. Cengel. "Thermodynamics an Engineering Approach", Tata McGraw-Hill Co. Ltd., 3rd Edition, 2002.

REFERENCES

1. Mayhew, A. and Rogers, B., "Engineering Thermodynamics", Longman Green & Co. Ltd., London, E.L.B.S. Edition, 1990.
2. Van Wylen, G.J. and Sonntag, R.E., "Fundamentals of Classical Thermodynamics (S.I. Version)", Second Edition, 1986.
3. Bacon, D.H., "Engineering Thermodynamics", Butterworth & Co., London, 1989.
4. Saad, M.A., "Thermodynamics for Engineers", Prentice-Hall of India Pvt. Ltd., 1989.
5. Reynolds, "Thermodynamics", Int. Student Edn., McGraw-Hill Book Co., Ltd., 1990

LIST OF EXPERIMENTS

1. Performance test on a 4-stroke engine
2. Valve timing of a 4 - stroke engine and port timing of a 2 stroke engine
3. Determination of effectiveness of a parallel flow heat exchanger
4. Determination of effectiveness of a counter flow heat exchanger
5. Determination of the viscosity coefficient of a given liquid
6. COP test on a vapour compression refrigeration test rig
7. COP test on a vapour compression air-conditioning test rig
8. Study of a Gas Turbine Engine.
9. Determination of Conductive Heat Transfer Coefficient.
10. Determination of Thermal Resistance of a Composite wall.

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Aircraft Maintenance Engineering, III Semester

AF 304 - FLUID MECHANICS AND MACHINERY

GOAL

To introduce the behaviour of fluids, kinematics and dynamics of fluids and hydraulic Machines

OBJECTIVES

The course should enable the students to :

1. Understand the principles of Basic concepts and properties of Fluid
2. Understand the Fluid Kinematics and its Dynamics
3. Study the basic concepts of Incompressible Flows
4. Study the basic concepts of Fluid Machines and Hydraulic turbines
5. To study the Hydraulic pumps & its applications

OUTCOME

The students should be able to :

1. The basic terms like Pressure, Density, Surface Tension & Fluid Statics
2. The types of flows, stream functions, Velocity Potential & familiarize in equations of Fluid Motion
3. The Laminar Flows, Flow through Pipes, Boundary Layers
4. The working Principles of Various Turbines like Kaplan, Pelton, Francis
5. The working Principles of Pumps like Centrifugal & Reciprocating Pumps

UNIT I BASIC CONCEPTS AND PROPERTIES

Fluid - definition, distinction between solid and fluid - Units and dimensions - Properties of fluids - density, specific weight, specific volume, specific gravity, temperature, viscosity, compressibility, vapour pressure, capillary and surface tension - Fluid statics: concept of fluid static pressure, absolute and gauge pressures - pressure measurements by manometers and pressure gauges.

UNIT II FLUID KINEMATICS AND FLUID DYNAMICS

Fluid Kinematics - Flow visualization - lines of flow - types of flow - velocity field and acceleration - continuity equation (one and three dimensional differential forms). Equation of streamline - stream function - velocity potential function - circulation - flow net. Fluid dynamics - equations of motion - Euler's equation along a streamline - Bernoulli's equation - applications - Venturi meter, Orifice meter, Pitot tube - dimensional analysis - Buckingham's pi theorem- applications - similarity laws and models.

UNIT III INCOMPRESSIBLE FLUID FLOW

Viscous flow - Navier-Stoke's equation (Statement only) - Shear stress, pressure gradient relationship - laminar flow between parallel plates - Laminar flow through circular tubes (Hagen poiseuille's) -

Hydraulic and energy gradient - flow through pipes - Darcy - weisback's equation - pipe roughness - friction factor- Moody's diagram-minor losses - flow through pipes in series and in parallel - power transmission - Boundary layer flows, boundary layer thickness, boundary layer separation - drag and lift coefficients.

UNIT IV HYDRAULIC TURBINES

Fluid machines: definition and classification - exchange of energy - Euler's equation for turbo machines - Construction of velocity vector diagram's - head and specific work - components of energy transfer - degree of reaction.

Hydro turbines: definition and classifications - Pelton turbine - Francis turbine - propeller turbine - Kaplan turbine - working principles - velocity triangles - work done - specific speed - efficiencies - performance curve for turbines.

UNIT V HYDRAULIC PUMPS

Pumps: definition and classifications - Centrifugal pump: Classifications, working principles, velocity triangles, specific speed, efficiency and performance curves. Reciprocating pump: classification, working principles, indicator diagram, work saved by air vessels and performance curves - cavitations in pumps - rotary pumps, working principles of gear and vane pumps.

TEXT BOOK

1. Streeter, V.L., and Wylie, E.B., "Fluid Mechanics", McGraw-Hill, 1983.
2. Kumar, K.L., "Engineering Fluid Mechanics", Eurasia Publishing House (P) Ltd., New Delhi (7th edition), 1995.
3. Vasandani, V.P., "Hydraulic Machines - Theory and Design", Khanna Publishers, 1992.

REFERENCES

1. Bansal, R.K., "Fluid Mechanics and Hydraulics Machines", (5th edition), Laxmi publications (P) Ltd., New Delhi, 1995.
2. White, F.M., "Fluid Mechanics", Tata McGraw-Hill, 5th Edition, New Delhi, 2003.
3. Ramamirtham, S., "Fluid Mechanics and Hydraulics and Fluid Machines", Dhanpat Rai and Sons, Delhi, 1991.
4. Som, S.K., and Biswas, G., "Introduction to fluid mechanics and fluid machines", Tata McGraw-Hill, 2nd edition, 2004.

LIST OF EXPERIMENTS

1. Calibration of venturimeter
2. Pressure measurement with Pitot static tube
3. Determination of pipe flow losses.
4. Verification of Bernoulli's theorem
5. Flow visualization by Heleshaw apparatus
6. Performance test on centrifugal pumps
7. Performance test on reciprocating pumps
8. Performance test on pelton wheel turbine
9. Performance test on Francis turbine
10. Determination of Viscosity of a Fluid

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Aircraft Maintenance Engineering, III Semester

AF 305 - AIRCRAFT MATERIALS

GOAL

To introduce various materials used in Aerospace industry, their behavior and testing methods

OBJECTIVES

The course should enable the students to :

1. Know about various types of materials and Knowledge of various types of hardness testing machines and various types of hardness numbers Linear and non-linear elastic properties- Stress and Strain Curves.
2. Know about the materials used in aircraft construction- Aluminium, Magnesium and Titanium
3. Know about the materials used in aircraft construction- Steel, Copper alloys and Super alloys.
4. Know about the adhesives and sealants used in aircraft industries.
5. Know about the non metals used in aircraft construction.

OUTCOME

The students should be able to :

1. Understand the different materials used and know the various types of hardness testing machine. Knowledge of Stress-strain curves for different type of materials.
2. Acquire knowledge about the properties of the material, the process of machining them and heat treating them.
3. Acquire knowledge about the specification of materials, their structural applications and properties.
4. Find out the different types of adhesives and sealant used, their advantages and the knowledge of the sandwich and honeycomb structure.
5. Acquire knowledge about the non metals like wood, fabrics, glass, plastics and the use of composite materials.

UNIT I MECHANICAL BEHAVIOUR OF ENGINEERING MATERIALS

Knowledge of various types of hardness testing machines and various types of hardness numbers Linear and non-linear elastic properties - Stress and Strain Curves - Yielding and strain Hardening , Toughness - Modules of resilience -- Baugher's effect - Effect of notches - Testing and flaw detection of materials and components.

UNIT II MATERIALS IN AIRCRAFT CONSTRUCTION - I

Aluminium and its alloys: Types and identification. Properties - Castings - Heat treatment processes - Surface treatments.

Magnesium and its alloys: Cast and Wrought alloys - Aircraft application, features specification, fabrication problems, Special treatments.

Titanium and its alloys: Applications, machining, forming, welding and heat treatment.

UNIT III MATERIALS IN AIRCRAFT CONSTRUCTION - II

Steels : Plain and low carbon steels, various low alloy steels, aircraft steel specifications ,corrosion and heat resistant steels, structural applications .

Maraging Steels: Properties and Applications

Copper Alloys - Monel, K Monel

Super Alloys: Use - Nickel base - Cobalt base - Iron base - Forging and Casting of Super alloys - Welding, Heat treatment.

UNIT IV ADHESIVE AND SEALANTS FOR AIRCRAFT

Advantages of Bonded structure in airframes - Crack arresting - Weight saving - Technology of adhesive Bonding Structural adhesive materials - Test for bonding structure

Typical bonded joints & non destructive tests for bonded joint

Bonded Sandwich structures - Materials - Methods of construction of honeycombs

UNIT V NON METALS IN AIRCRAFT CONSTRUCTION

Wood and fabric in aircraft construction and specifications -Glues Use of glass, plastics and rubber in aircraft, Introduction to glass and carbon composite

TEXT BOOKS

1. Lalith Gupta, "Aircraft General Engineering" Himalaya Book House, Delhi 2003
2. Hajira Chowdhry, "Workshop Technology " - Vol 1 & 2 ,Nedia Promoters, Mumbai

REFERENCE

1. "Aircraft Material & Process", Titterton 2004
2. "Advanced Composite Materials", Lalith Gupta 2006, Himalaya Book House, Delhi.
3. "Advanced Composite Materials", Syney Foreman 2006, Tata Macgrahill.

List of experiments :

1. Metallographic studies – Study of Optical microscope, Optically flat surface preparation, etching reagents, Grain size- ASME no., micro structures, Image analysis, Standard specimen,
2. Carbon, sulphur, Phosphorus determination, Strauhlin's apparatus, Eggert's Method in different samples.
3. Hardness and Hardenability test, Jeremy Cony test. Soft and hard Martensite.
4. Different heat treatment cycles using electric furnace [Programmable preferred], Annealing, Case carburising, Normalising, etc.
5. Gravimetric / Volumetric - chemical analysis of alloying elements like, Cr, Ni, Mn, Si etc.
6. Study of different instrumental method of analysis, spectrophotometers, Differential Scanning calorimeter,
7. Spot test for quick assessment of alloying elements like Mn, Cr, Ni, etc.
8. Experiments / study of Non Destructive Methods, Ultrasonic test, Magnetic particle inspection, Dye penetration test, Eddy current test, Radiography test.

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Aircraft Maintenance Engineering, III Semester

AF-306 DESIGN AND DRAFTING LAB

LIST OF EXERCISES

1. Design of riveted joints (Lap joint)
2. Design of riveted joints (Butt joint with single and double straps).
3. Design of welded joints.
4. Layout of typical wing structure.
5. Layout of typical fuselage structure.
6. Computer aided modelling of typical aircraft wing.
7. Computer aided modelling of typical fuselage structure.
8. Computer aided modelling of landing gear
9. Three view diagram of a typical aircraft
10. Layout of control systems